

ANANYA REETHA NOBLE

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RESEARCH INTERESTS

Focused on large language models, parameter-efficient fine-tuning, and retrieval-augmented generation, with applied experience in computer vision and deep learning. Interested in building efficient, deployable AI systems that bridge foundational research and real-world impact.

EDUCATION

Defence Institute of Advanced Technology

M. Tech in Computer Science and Engineering (Artificial Intelligence)

Expected Graduation: May 2027

Relevant Courses: Responsible AI, Practical Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, Adversarial and Generative AI, Large Language Models

Girinagar, Pune

Current CGPA: 9.05

Mar Athanasius College of Engineering (APJ Abdul Kalam Technological University)

B. Tech in Computer Science and Engineering

Graduation: May 2024

Kothamangalam, Kerala

CGPA: 9.25

PUBLICATIONS

Exploring LoRA for Parameter-Efficient Fine-tuning of LLMs in Enhanced Algorithm-to-Python-Source-Code Translation Task

Authors: Allwyn Bat Thomas, Ananya Reetha Noble, Anna Wilson, Leya Elizabeth Sunny, Rini Thazhathoot Paul

Conference: ITechCET 2024

Published in AIP Conference Proceedings

DOI: <https://doi.org/10.1063/5.0247544>

Manuscript in Preparation:

Comparative Analysis of Baseline and Custom DL Models for Biofouling Detection and Classification

Authors: Mohineet Kaur, Smruthi Lakadwala, Ananya Reetha Noble, Divya Sharma, Akash Kapoor, Sunita Dhavale

RESEARCH PROJECTS

Exploring LoRA for Parameter-Efficient Fine-Tuning of LLMs in Algorithm-to-Python Source Code Translation

- Investigated parameter-efficient adaptation of LLMs for automated translation of algorithmic descriptions into executable Python code.
- Fine-tuned Mistral 7B using LoRA (rank=32, 4-bit quantization) on a custom-constructed dataset of 18,000 algorithm-code pairs, publicly released on Hugging Face. ([Dataset](#))
- Achieved 93.5% accuracy and perplexity of 2.03, outperforming prior approaches including Seq2Seq (88.7%).
- Published and presented findings at ITechCET 2024, later featured in AIP Conference Proceedings. ([DOI](#))

Comparative Analysis of Baseline and Custom DL Models for Biofouling Detection and Classification

- Developed a deep learning framework for six-class marine biofouling classification under simulated underwater imaging conditions.
- Built a custom underwater degradation augmentation pipeline (colour attenuation, turbidity, backscatter, non-uniform illumination) to expand a curated 448-image dataset to 3,443 training samples.
- Systematically evaluated seven model variants across EfficientNetV2, DenseNet121, and YOLOv8m, incorporating CBAM attention and BYOL self-supervised pretraining.
- The proposed EfficientNetV2-S with CBAM and BYOL pretraining achieved a validation mAP@50 of 0.9624 and macro F1-score of 0.94, suitable for deployment on autonomous underwater vehicles.

Feature Selection and Salary Prediction using Rough Set Theory ([Code](#))

- Applied Fuzzy Rough Set Theory with the Quick Reduct algorithm for feature selection on Stack Overflow Developer Survey data (2018–2020), identifying the minimal subset of attributes, including education, location, and years of experience, most predictive of salary.
- Trained a Decision Tree regression model on the selected features, achieving R^2 scores of up to 0.9997 and MAE as low as 153.32 across three yearly datasets, with data preprocessing handled via Scikit-learn and Pandas.

OTHER PROJECTS

Local RAG Chatbot with Document QA ([Code](#))

Built a fully offline retrieval-augmented generation (RAG) chatbot using Gemma 2B via Ollama, LangChain, ChromaDB vector search, and sentence-transformers; deployed via Docker with a FastAPI backend supporting PDF/TXT document upload and context-grounded question answering.

Deep Learning Implementations from Scratch ([Code](#))

Implemented core deep learning architectures from scratch in PyTorch including MLP, CNN, VGG19, DenseNet, and InceptionV3 on CIFAR-10/MNIST, and an RNN for Netflix stock price forecasting.

Air Quality Prediction Model for PM2.5 Levels in India ([Code](#))

Built a regression model to predict PM2.5 pollutant levels from 36,000+ hourly air quality records (2017–2022). Engineered temporal features from raw timestamps, applied Label Encoding for categorical time bins, and compared Linear Regression, Decision Tree, and Random Forest regressors with GridSearchCV hyperparameter tuning, achieving an RMSE of 8.97 with the optimized Decision Tree model.

SKILLS

Technical Writing, Documentation, Community Building

Programming: Python, C, Java, SQL

ML/AI Frameworks: PyTorch, Hugging Face Transformers, LoRA, PEFT, Scikit-learn, Pandas, LangChain, ChromaDB, Sentence Transformers, YOLOv8, BYOL

Tools: Git, GitHub, Docker, FastAPI, Google Colab, Unix/Linux, VSCode

Techniques: LLMs, Parameter-Efficient Fine-Tuning, Retrieval-Augmented Generation (RAG), Transfer Learning, Computer Vision, Self-Supervised Learning

CERTIFICATIONS

Python for Data Science (IBM)

LEADERSHIP AND VOLUNTEERING EXPERIENCE

- Co-authored and presented research at ITechCET 2024 (AIP Conference Proceedings), contributing to the academic discourse on parameter-efficient fine-tuning of LLMs.
- Contributed to open-source research infrastructure by publicly releasing a custom 18,000-sample algorithm-code pairs dataset on Hugging Face, supporting reproducibility and community research.
- Organized and coordinated technical content and knowledge-sharing efforts across student-led engineering communities (NetX Club, IEEE Kochi Hub, IEEE MACE SB), fostering collaborative learning in AI and computer science.